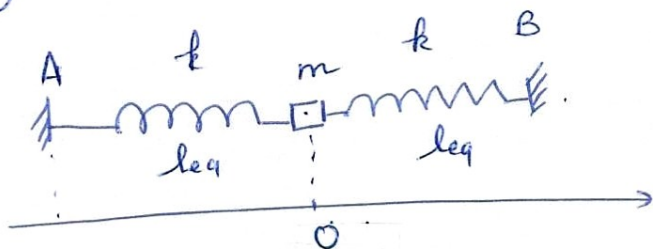
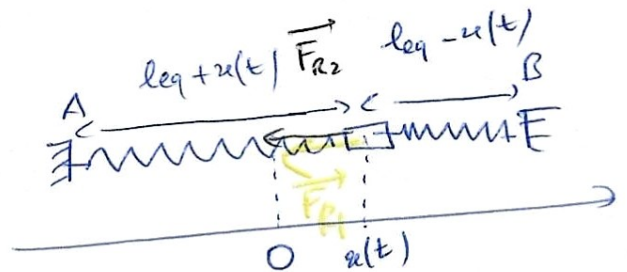
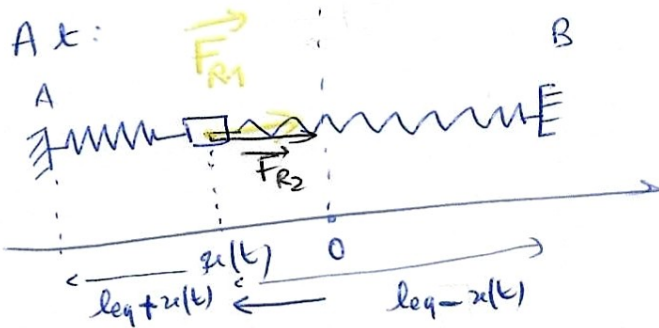


# Oscillateur à 2 ressorts.

① Au repos :



$l_0$ .



②  $\vec{F}_{R1} = -k(l_0 + x(t) - l_0) \vec{u}_x$

③  $\vec{F}_{R2} = k(l_0 - x(t) - l_0) \vec{u}_x$

④ Bilan des forces exercées sur m :

- tension du ressort 1  $\vec{F}_{R1}$
- " 2  $\vec{F}_{R2}$
- poids  $\vec{P}$
- réaction de la tige  $\vec{R}$ .

④ Le mouvement étant purement horizontal,  $\vec{P} + \vec{R} = \vec{0}$ .  
 PFD à la masse ds le ref du labo, supposé galiléen.

$$\Sigma \vec{F} = m \vec{a}$$

$$\vec{F}_{R1} + \vec{F}_{R2} + \vec{P} + \vec{R} = m \vec{a}$$

$$-k(l_0 + x(t) - l_0) + k(l_0 - x(t) - l_0) = m \ddot{x} \quad \text{projection sur } \vec{u}_x$$

$$-2kx(t) = m \ddot{x}$$

$$-\frac{2k}{m} x(t) = \ddot{x}$$

$$\Rightarrow \ddot{x} + \frac{2k}{m} x(t) = 0$$

⑤ On reconnaît 1 oscillateur harmonique de pulsation

$$\omega_0 = \sqrt{\frac{2k}{m}}$$

⑥

$$T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0} = 2\pi \sqrt{\frac{m}{2k}} = \pi \sqrt{\frac{2m}{k}}$$

⑧ La solution est de la forme :

$$u(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

⑦

$$CJ: \begin{cases} u(0) = u_0 \\ \dot{u}(0) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} u(0) = A = u_0 \\ \dot{u}(0) = B\omega_0 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} A = u_0 \\ B = 0 \end{cases}$$

$$\dot{u}(t) = -\omega_0 A \sin(\omega_0 t) + B\omega_0 \cos(\omega_0 t)$$

$$\dot{u}(0) = B\omega_0$$

Finalement

$$u(t) = u_0 \cos(\omega_0 t)$$

$$\dot{u}(t) = -\omega_0 u_0 \sin(\omega_0 t)$$

$$E_p(t) = E_p^1(t) + E_p^2(t)$$

$$= \frac{1}{2} k (l_{eq} + u(t) - l_0)^2 + \frac{1}{2} k (l_{eq} - u(t) - l_0)^2 + K$$

⑨

$$E_c(t) = \frac{1}{2} m \dot{u}^2(t)$$

$$= \frac{1}{2} m \omega_0^2 u_0^2 \sin^2(\omega_0 t)$$

$$= \frac{1}{2} m \cdot \frac{2k}{m} u_0^2 \sin^2(\omega_0 t)$$

$$E_c(t) = k u_0^2 \sin^2(\omega_0 t)$$

$$\left( l_{eq} - l_0 + x(t) \right)^2 + \left( l_{eq} - l_0 - x(t) \right)^2$$

$$2(l_{eq} - l_0)^2 + 2x(t)^2 + \cancel{2(l_{eq} - l_0)x(t)} - \cancel{2(l_{eq} - l_0)x(t)}$$

⑩

$$E_p(t) = \frac{1}{2} k \cdot 2x^2(t) + \underbrace{\frac{1}{2} k \cdot 2(l - l_0)^2 + K}_{=0}$$

$$= kx^2(t)$$

$$E_p(t) = kx_0^2 \cos^2(\omega_0 t)$$

$$\omega_0^2 = \frac{2k}{m}$$

⑪ Analogie :  $E_m(t) = E_c(t) + E_p(t)$

$$E_m(t) = kx_0^2 = \text{cte}$$

⑫ L'énergie mécanique est conservée puisqu'on a réglé les frotts.

14.

$$\vec{F}_{R1} + \vec{F}_{R2} + \vec{P} + \vec{R} + \vec{F} = m\vec{a}$$

$$-k(l_{eq} + u(t) - l_0) + k(l_{eq} - u(t) - l_0) - \mu \ddot{u} = m \ddot{u} \text{ projecté selon } O x$$

$$-2k u(t) - \mu \ddot{u}(t) = m \ddot{u}$$

$$-\frac{2k}{m} u(t) - \frac{\mu}{m} \ddot{u}(t) = \ddot{u}$$

$$\ddot{u} + \frac{\mu}{m} \ddot{u}(t) + \frac{2k}{m} u(t) = 0$$

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$$\ddot{u} + h \ddot{u}(t) + \omega_0^2 u(t) = 0$$

$$\omega_0 = \sqrt{\frac{2k}{m}}$$

$$h = \frac{\mu}{m}$$

Equation d'un oscillateur amorti.

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$$\Delta = h^2 - 4\omega_0^2$$

$$= h^2 - 4 \cdot \frac{2k}{m}$$

$$\Delta = h^2 - \frac{8k}{m}$$

L'éq se réécrit  $\ddot{u} + \frac{\omega_0}{\Phi} \ddot{u}(t) + \omega_0^2 u(t)$

avec  $\frac{\omega_0}{\Phi} = h \Rightarrow \Phi = \frac{\omega_0}{h} = \frac{1}{h} \sqrt{\frac{2k}{m}}$

On a 1 régime pseudo-périodique

$$\Leftrightarrow \Phi > 1/2$$

$$\Leftrightarrow \frac{1}{h} \sqrt{\frac{2k}{m}} > 1/2$$

$$\Leftrightarrow \frac{m}{\mu} \sqrt{\frac{2k}{m}} > 1/2$$

$$\Leftrightarrow \frac{1}{\mu} \sqrt{2mk} > 1/2$$

$$\Leftrightarrow 2\sqrt{2mk} > \mu$$

$$\Leftrightarrow \mu < 2^{3/2} \sqrt{k m}$$

17 On a obtenu :

$$u(t) = e^{-\frac{\omega_0}{2Q}t} \left( A \cos(\Omega t) + B \sin(\Omega t) \right)$$

19 Appliquons les CI

$$\begin{aligned} \dot{u}(t) &= -\frac{\omega_0}{2Q} e^{-\frac{\omega_0}{2Q}t} (A \cos \Omega t + B \sin \Omega t) \\ &+ e^{-\frac{\omega_0}{2Q}t} (-A\Omega \sin \Omega t + B\Omega \cos \Omega t). \end{aligned}$$

On a

$$\begin{cases} \dot{u}(0) = 0 \\ u(0) = u_0 \end{cases}$$

ce qui donne

$$\begin{cases} -\frac{A\omega_0}{2Q} + B\Omega = 0 \\ A = u_0 \end{cases}$$

ce qui donne

$$\begin{cases} A = u_0 \\ B = \frac{A\omega_0}{2Q\Omega} = \frac{\omega_0}{2Q\Omega} u_0 \end{cases}$$

d'où la solution :

$$u(t) = e^{-\frac{\omega_0}{2Q}t} \left( u_0 \cos \Omega t + u_0 \frac{\omega_0}{2Q\Omega} \sin \Omega t \right)$$

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$$\Omega = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}$$

$$\Omega = \omega_0 \sqrt{1 - \frac{h^2}{4\omega_0^2}}$$

$$Q = \frac{\omega_0}{h} \quad \frac{1}{Q} = \frac{h}{\omega_0}$$